

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 25

Question Count: 5

Code of Conduct

I **LEO LEENAS A (192511068)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **LEO LEENAS A (192511068)**

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Question 1: Handling Seasonal Traffic Spikes with Auto-Scaling and Load Balancing

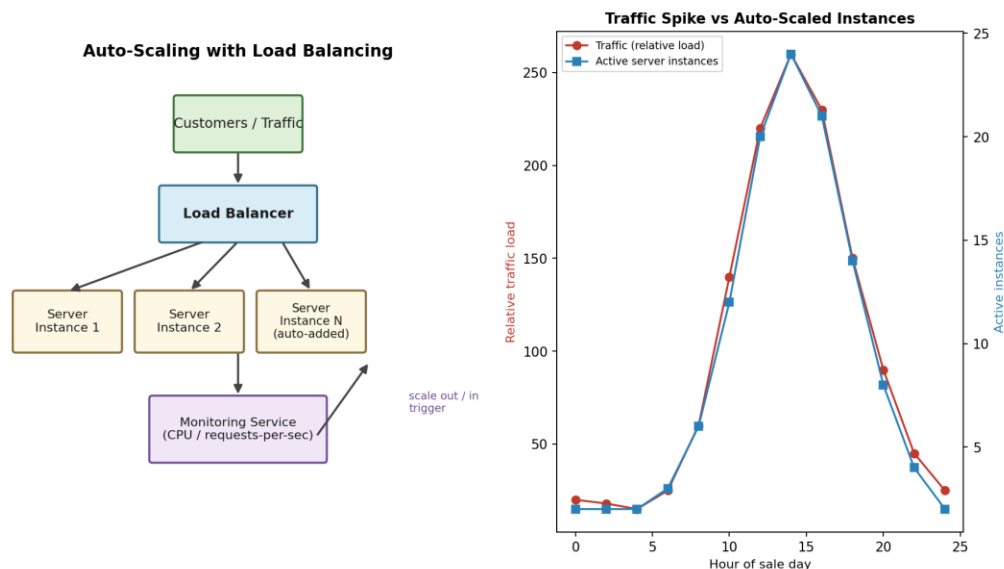
Scenario — A company experiences unpredictable traffic spikes during seasonal sales, and needs a way to handle the demand without the site slowing down or crashing.

When an e-commerce site runs a sale, the number of visitors can jump from a few thousand to several hundred thousand within an hour. A fixed set of servers sized for normal days will simply choke under this kind of load. Cloud auto-scaling solves this by watching resource usage in real time and adding new server instances automatically whenever CPU usage, memory, or the number of incoming requests crosses a set limit. Once the sale traffic drops back down, the extra instances are removed automatically, so the company is not paying for capacity it does not need for the rest of the year.

Load balancing works alongside auto-scaling. Instead of every customer request going to one server, the load balancer spreads the traffic evenly across all the active instances. This does two things at once — it stops any single server from becoming a bottleneck, and if one instance fails or becomes unhealthy, the load balancer simply stops sending traffic to it and reroutes the requests to the healthy ones. Because of this, users rarely notice a slowdown even though the backend is scaling up and down behind the scenes. Downtime is reduced because there is no single point of failure, and the checkout process, product pages, and payment gateway keep responding quickly even at peak load.

Monitoring tools are what make the whole setup intelligent rather than reactive. Services such as Amazon CloudWatch or Azure Monitor continuously track metrics like CPU utilization, request latency, and error rates, and they trigger the auto-scaling policy the moment a threshold is crossed. Without monitoring, the system would have no way of knowing that traffic is rising until customers start complaining about a slow site. Monitoring also gives the operations team dashboards and alerts, so if something unusual happens, like a sudden error spike from a payment gateway, someone can step in before it becomes a bigger issue.

A good real-world example is Flipkart's Big Billion Days sale or Amazon's Prime Day. Both platforms see traffic that is many times their normal load within minutes of the sale opening. Their infrastructure is built on cloud auto-scaling groups behind load balancers, with monitoring dashboards tracking order rates and server health every second. This is exactly why these platforms can process millions of orders in a short sale window without the site going down, while a traditional, fixed-capacity data center would almost certainly fail under the same load.



Auto-scaling architecture with load balancer, dynamic instances and monitoring feedback, alongside a sample traffic-vs-instance-count curve for a sale day

Question 2: Durability, Redundancy and Cost Advantages of Cloud Backup Storage

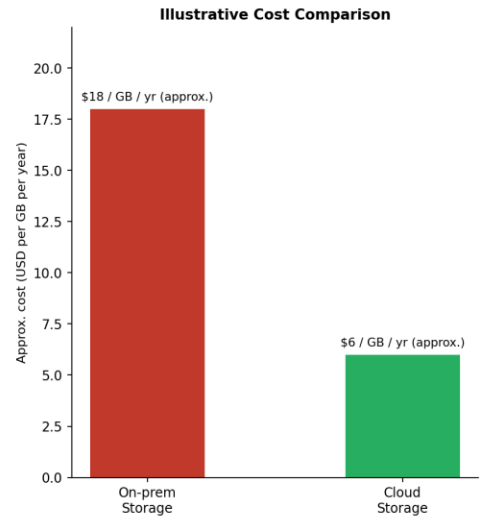
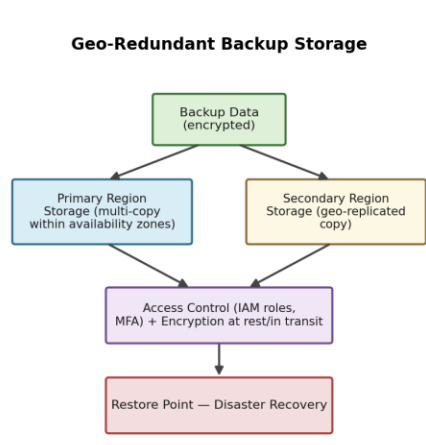
Scenario — An organization wants to store large volumes of backup data securely in the cloud and needs assurance that the data will not be lost.

Cloud storage providers achieve durability by never keeping a single copy of any file. When data is uploaded, it is automatically split and copied across multiple physical disks, and usually across more than one data center within a region. Amazon S3, for example, advertises eleven nines of durability, meaning the chance of losing an object in a given year is close to zero, because the same object exists on several independent storage devices at once. Some providers also use erasure coding, which breaks data into fragments and adds parity information, so the original file can be rebuilt even if a few fragments are lost. On top of this, most cloud storage classes support cross-region replication, so an entire copy of the backup can sit in a second geographic region in case the first region has an outage.

Encryption and access control are what keep this backup data safe from unauthorized access rather than just accidental loss. Data is encrypted at rest using something like AES-256, so even if a physical disk were somehow removed from the data center, the contents would be unreadable without the key. Data is also encrypted in transit using TLS while it moves between the client and the cloud, so it cannot be intercepted on the network. Access control is handled through identity and access management (IAM) policies, where only specific users or applications are given permission to read or write the backup, and multi-factor authentication adds another layer so that a stolen password alone is not enough to reach the data.

The cost advantage compared to a traditional on-premise backup setup is significant. A company running its own tape library or disk array has to buy the hardware upfront, keep spare capacity for growth, pay for a physically secured room, and maintain staff to manage it. Cloud backup storage instead works on a pay-as-you-use model — the organization is billed only for the storage actually consumed, and it can shrink or expand instantly without buying new hardware. There is also no need to plan years in advance for capacity, since the cloud provider handles that scaling internally.

A practical disaster recovery example: suppose a company's main office data center suffers a fire or a flood and the local servers are destroyed. Because the backup data was already replicated to a cloud region hundreds of kilometers away, the company can restore its databases and files from that cloud copy within hours instead of losing years of records permanently. This is the core reason most organizations now treat cloud backup as their last line of defense rather than an optional extra.



Cloud storage removes upfront hardware cost and follows a pay-as-you-use model

Geo-redundant backup storage with encryption and access control, and an approximate cost comparison between on-premise and cloud storage

Question 3: Improving Global Performance with CDNs and Edge Servers

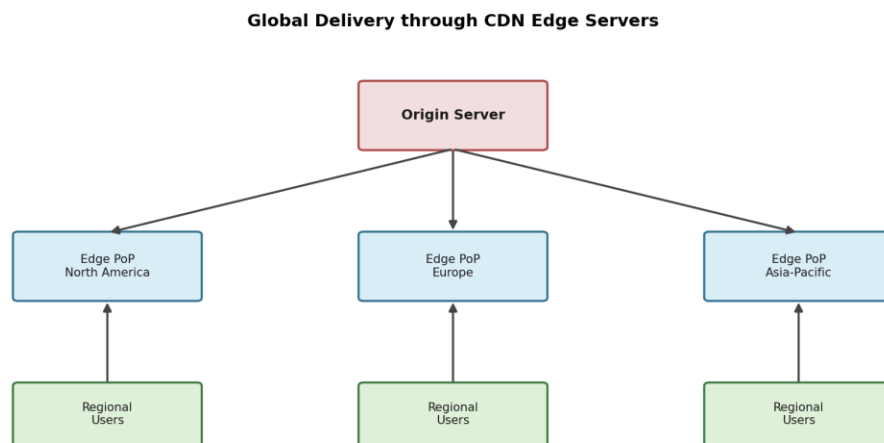
Scenario — A mobile app company plans to deploy its application globally and needs consistent performance for users in different regions.

If every user request had to travel back to one central server, users far away from that server would experience noticeably higher loading times, simply because of the physical distance the data has to travel. A content delivery network solves this by caching copies of static content, such as images, videos, app assets, and even some API responses, on servers placed in many different geographic locations. When a user in Tokyo opens the app, the request is served from a server in or near Tokyo instead of one in, say, the United States. This cuts down the round-trip time dramatically and is the main reason CDNs reduce latency so effectively.

Edge servers take this idea a step further. Rather than only storing cached copies, edge locations can also run small pieces of application logic close to the user, such as authentication checks, image resizing, or personalization rules. This means the user's device does not need to wait for a distant origin server to process every request, and the overall experience feels far more responsive, especially on mobile networks where latency is already a bigger factor than it is on broadband. For regions with weaker connectivity, this difference is even more noticeable, because every extra network hop is more costly in terms of delay.

Cloud providers support global availability by operating in multiple regions and availability zones around the world, and by pairing this with their CDN offerings, such as AWS CloudFront, Google Cloud CDN, or Azure CDN. A company does not need to build its own physical infrastructure in each country. It simply configures its application to use the provider's global edge network, and the provider takes care of routing each user to the nearest healthy edge location, along with failover if one location goes down.

This is very visible in streaming and gaming applications. A platform like Netflix places its content on edge caches close to internet service providers around the world, so a movie starts playing almost instantly instead of buffering. Mobile multiplayer games such as PUBG Mobile or Call of Duty Mobile rely on regional edge servers to keep player-to-server latency low, because even a delay of 100 to 150 milliseconds can be the difference between winning and losing a match. Both cases show why CDN and edge deployment is treated as a core part of any application meant to be used globally, not just an optional performance upgrade.



Content is cached close to each user, so requests no longer travel back to a single distant origin server

CDN edge servers caching content close to regional users instead of routing every request to a single origin server

Question 4: Regulatory Compliance and Data Security for Financial Cloud Usage

Scenario — A financial institution requires strict compliance and data security in cloud usage while handling sensitive customer transactions.

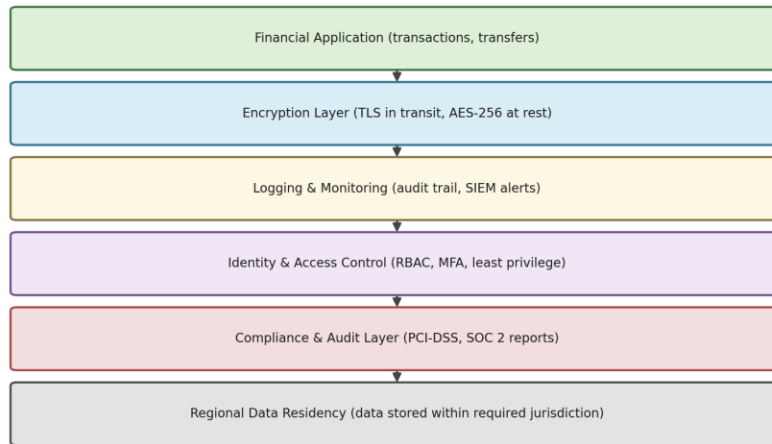
Cloud providers support regulatory compliance mainly through certifications and a shared responsibility model. Providers such as AWS, Microsoft Azure, and Google Cloud undergo independent audits against standards like PCI-DSS for payment data, ISO 27001 for information security, and SOC 2 for controls around availability and confidentiality. This gives the financial institution documented proof that the underlying infrastructure meets industry standards, but the institution is still responsible for configuring its own applications, access policies, and data handling correctly on top of that infrastructure — this split of responsibility is exactly what the shared responsibility model describes.

Logging, monitoring, and encryption form the backbone of day-to-day security in this setup. Every action taken on cloud resources, from a login attempt to a change in a database configuration, can be logged through services like AWS CloudTrail or Azure Activity Log, creating an audit trail that regulators can review later if needed. Monitoring tools sit on top of these logs and raise alerts when something abnormal happens, such as a login from an unexpected country or a sudden spike in failed transactions, which could indicate fraud or a breach in progress. Encryption protects the transaction data itself, both while it is stored in the database and while it moves between systems, so that even if a log or a database were somehow accessed without authorization, the actual account numbers and transaction details would remain unreadable.

Data sovereignty is handled by choosing where the data physically lives. Most cloud providers let a customer pin their storage and processing to a specific country or region, and some offer dedicated sovereign cloud regions built specifically for government and financial customers. A bank operating in India, for instance, can configure its cloud resources so that all customer transaction data stays within data centers located in India, satisfying RBI's data localization requirements, while still using the provider's global services for things like analytics that do not involve sensitive personal data.

A concrete example would be a bank processing UPI or card transactions through a cloud-hosted payment gateway. The transaction data is encrypted the moment it is captured, every access to that data is logged for audit purposes, the servers processing it are located within the country as required by the regulator, and automated monitoring flags anything that looks like fraudulent activity in near real time. This combination is what allows a regulated financial institution to use public cloud infrastructure while still passing the compliance audits it is legally required to undergo.

Compliance and Security Layers for Financial Data



Layered security and compliance controls for financial transaction data, from encryption through to regional data residency

Question 5: Containerization, Orchestration and Portability for Microservices

Scenario — A development team wants to adopt containerization for application deployment instead of relying purely on virtual machines.

The main difference between a container and a virtual machine comes down to what each one has to carry with it. A virtual machine includes a complete guest operating system on top of the hypervisor, so every VM is booting and running its own kernel, which takes up a noticeable amount of CPU, memory, and disk space, and typically takes tens of seconds or more to start. A container, on the other hand, shares the host machine's operating system kernel and only packages the application along with its libraries and dependencies. Because of this, containers are usually only a few megabytes to a few hundred megabytes in size compared to gigabytes for a VM, and they can start in a second or two. This makes it possible to run far more containers than VMs on the same physical hardware, which is a major reason companies are shifting toward containers for microservices.

As the number of containers grows, running them by hand becomes unmanageable, and that is the problem orchestration tools like Kubernetes are built to solve. Kubernetes takes care of scheduling containers onto available worker nodes, restarting them automatically if they crash, and scaling the number of running copies up or down based on load, without a human having to log into a server and issue commands manually. If a node in the cluster fails, Kubernetes reschedules the containers that were running on it onto healthy nodes, so the application keeps running with minimal disruption. This is what allows a small development team to run a fairly complex, multi-service application in production without needing a large dedicated operations team watching every server around the clock.

Portability is another strong point in favor of containers. Because a container packages the application together with everything it needs to run, the same container image will behave the same way whether it is running on a developer's laptop, an on-premise server, or any major cloud provider. This avoids the classic 'it worked on my machine' problem and means a company is not locked into one specific cloud vendor, since the same Docker image can be redeployed on AWS, Azure, Google Cloud, or a private data center with very little change to the deployment process.

A good example of this in practice is a food delivery application split into separate microservices — one service handling order placement, another handling payments, another handling delivery tracking, and another sending notifications. Each of these runs as its own container, can be updated and redeployed independently without touching the others, and Kubernetes scales the order service up during lunch and dinner rush hours while keeping the notification service at a smaller, steady size. This kind of independent, fine-grained scaling would be far harder to achieve if the whole application were still packaged as a small number of large virtual machines.

Virtual Machines

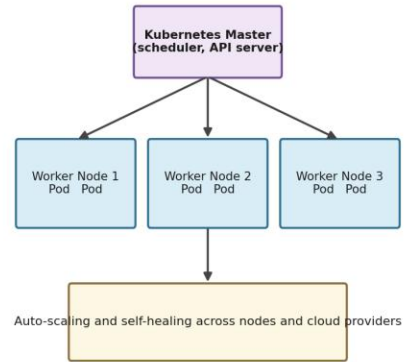


Containers



Containers share one OS kernel, so they start faster and use fewer resources than full VMs

Kubernetes Orchestration



Kubernetes keeps microservices running and scales pods automatically

Virtual machine stack compared with a container stack, and a Kubernetes cluster orchestrating pods across worker nodes

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Total Marks Awarded: